DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING
Power Electronics Laboratory

LAB REPORT

Design and Development of a DC-DC Buck Converter

for Fan Speed Control (24V, 6W) from a 48V Supply

Submitted By,

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1. Introduction

This report documents the complete design and development of a DC-DC Buck Converter intended to power a 24V, 6W DC fan from a 48V DC supply. The converter is required to operate the fan at two distinct speeds: full speed at 24V and half speed at approximately 12V. Speed control is achieved by adjusting the duty cycle of the pulse-width modulated (PWM) gate signal driving the power MOSFET.

The buck (step-down) converter topology was selected because the output voltage is lower than the input, allowing efficient, regulated power delivery with minimal component count. The switching frequency was set at 20 kHz — the boundary of audible range — to reduce core and switching losses while keeping the filter components to a practical size.

A 96Ω resistor is used as the load in simulation, accurately representing the 24V, 6W fan at full speed ($P = V^2/R \Rightarrow R = 96\Omega$). The inductor (L1 = 1.9–2.0 mH) and filter capacitor (C1 = 100μF) are designed to maintain continuous conduction mode (CCM) and a ripple voltage within acceptable limits.

Note on simulation supply voltage: The theoretical design targets a 48V input. However, since a 48V laboratory supply was not readily available at the time of simulation, the LTspice model was run with a scaled 30V supply at 50% duty cycle (VS = 15V) to verify converter behaviour and waveform quality. All design equations, component sizing, and the inductor magnetics design are performed for the full 48V specification.

Subject	Power Electronics — Lab Report
Converter Type	DC-DC Buck Converter
Application	24V DC Fan Speed Control (6W)
Input Voltage	48V DC
Output Voltage	24V DC (Full Speed) / ~12V (Half Speed)
Switching Frequency	20 kHz
Date	May 2026

2. Theoretical Background

2.1 Voltage Conversion Ratio

The ideal DC voltage conversion ratio for a buck converter operating in CCM is:

$$V_{out} = D \times V_{in}$$

where D is the duty cycle ($0 \le D \le 1$) of the PWM signal. For this design:

Parameter	Value
Full speed (24V out)	**D = 24/48 = 0.50 (50%)**
Half speed (≈12V out)	**D = 12/48 = 0.25 (25%)**

2.2 Critical Inductance for CCM

$$L_{min} = (V_{in} - V_{out}) \times D / (2 \times f_{sw} \times I_{load})$$

$$L_{min} = (48 - 24) \times 0.50 / (2 \times 20{,}000 \times 0.25) = 1.2\ \text{mH}$$

The designed inductance of 2.0 mH comfortably exceeds Lmin, ensuring CCM at rated load.

2.3 Output Voltage Ripple

$$\Delta V_{out} = V_{out} \times (1 - D) / (8 \times L \times C \times f_{sw}^2)$$

$$\Delta V_{out} = 24 \times 0.5 / (8 \times 2\times10^{-3} \times 100\times10^{-6} \times 400\times10^{6}) \approx 18.75\ \text{mV}\ \ (0.078\%)$$

The output voltage ripple is extremely small (≤0.1%), confirming excellent filtering by the chosen L and C values.

3. Circuit Design

3.1 Circuit Specifications

Parameter	Value / Specification
Input Voltage (Vin)	48 V DC
Output Voltage (Vout)	24 V DC (full speed) / ~12 V (half speed)
Output Power (Pout)	6 W (fan at full speed)
Load Resistance (R1)	96 Ω (24V, 6W fan equivalent)
Load Current (Iload)	0.25 A (full speed)
Duty Cycle — Full Speed	D = 0.50
Duty Cycle — Half Speed	D = 0.25
Switching Frequency (fsw)	20 kHz
Filter Capacitor (C1)	100 µF
Filter Inductor (L1)	1.9 – 2.0 mH (designed: 2.0 mH)
Allowable Current Ripple	30% of Iload

3.2 Component Selection

3.2.1 Power MOSFET — IRF3205

The IRF3205 N-channel power MOSFET was selected as the main switching element.

Parameter	Value	Significance	Status
VDS (max)	55V	Exceeds 48V input ✓	Safe
ID (max)	110A	Far exceeds 0.25A load ✓	Safe
RDS(on)	8 mΩ	Extremely low conduction loss ✓	Optimal
Gate threshold VGS(th)	2–4V	Compatible with TLP250 gate driver output (12V VGS) ✓	OK

3.2.2 Freewheeling Diode

The Schottky diode is used as the freewheeling (catch) diode.

Parameter	Value / Reason
Reverse voltage (VRRM)	45V — adequate for 48V rail with margin
Forward current (IF)	6A — far exceeds design current
Forward voltage (VF)	~0.45V — low loss Schottky type
Why Schottky?	Faster reverse recovery, lower VF — reduces switching loss

3.2.3 Gate Driver — TLP250 (Floating Supply, VGS = 12V)

The TLP250 optocoupler-based gate driver IC is used to drive the IRF3205 MOSFET. The TLP250 provides galvanic isolation between the low-voltage PWM control signal (from the Arduino/microcontroller) and the high-side power circuit, making it particularly suitable for floating gate drive applications where the MOSFET source is not referenced to ground.

Floating Gate Supply Configuration:

A dedicated floating 12V DC supply (VGS = 12V) is used to power the output stage of the TLP250. This supply is referenced to the MOSFET source pin (VS node), not to circuit ground. This floating arrangement ensures that the gate-to-source voltage (VGS) is always a clean 12V regardless of the switching node voltage (which swings between 0V and 48V), thereby guaranteeing reliable and fast turn-on and turn-off of the IRF3205 at every switching transition.

TLP250 Key Specifications:

Parameter	Value	Significance in This Design
Supply Voltage (VCC2)	10 — 35V	12V floating supply used — within range ✓
Gate-Source Voltage (VGS)	12V (floating)	Ensures full enhancement of IRF3205 (VGS(th) = 2–4V) ✓
Peak Output Current	±1.5 A	Fast MOSFET gate charging at 20 kHz
✓ Isolation Voltage	2500 Vrms	Protects MCU from 48V power rail ✓
Propagation Delay	≤ 0.5 μs	Negligible at 20 kHz (period = 50 μs)
✓		
Input LED Current (IF)	5 — 16 mA typical	Driven directly from Arduino PWM pin via series resistor ✓

Operating Principle: When the Arduino PWM output goes HIGH, current flows through the TLP250 input LED. The optocoupler's phototransistor activates, and the TLP250 output stage (powered by the 12V floating supply) drives the MOSFET gate HIGH at 12V above the source (VGS = +12V), turning the MOSFET fully ON (RDS(on) = 8 mΩ). When the PWM output goes LOW, the LED switches off, the output pulls the gate LOW (VGS = 0V), and the MOSFET turns OFF, allowing the freewheeling diode to conduct. The duty cycle of the Arduino PWM signal (50% for full speed, 25% for half speed) directly controls the output voltage of the converter.

3.2.4 Input Filter Capacitor — C2 (100μF)

A 100μF input bypass capacitor (C2) is placed across the supply to suppress high-frequency switching noise and prevent voltage spikes from propagating back to the 48V source. It also acts as a local charge reservoir, reducing input current ripple.

3.3 Circuit Schematic

The complete buck converter circuit simulated in LTspice is shown below. Due to laboratory supply constraints, the simulation was conducted at 30V input (50% duty cycle, VS ≈15V) to verify waveform quality and converter dynamics. All design parameters are specified for the 48V system.

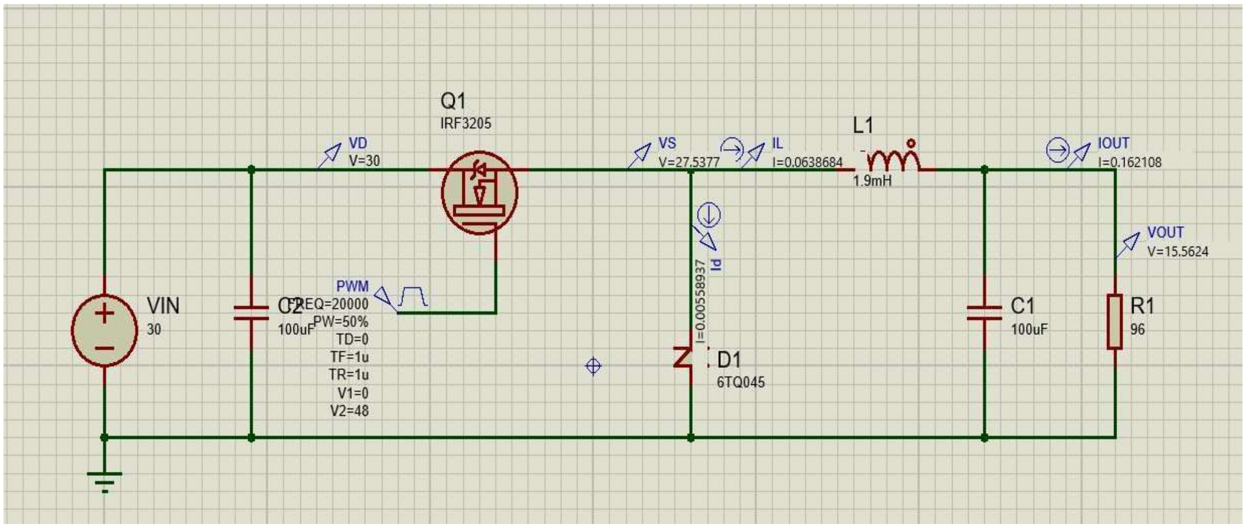


Figure 1: Proteus Simulation Schematic — DC-DC Buck Converter

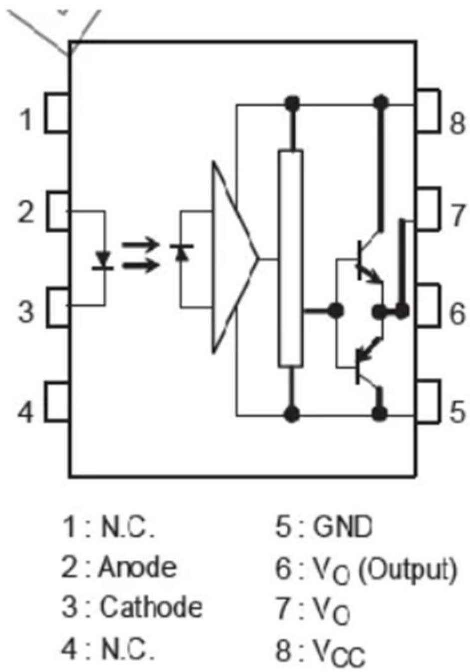


Figure 2: TLP250H circuit diagram

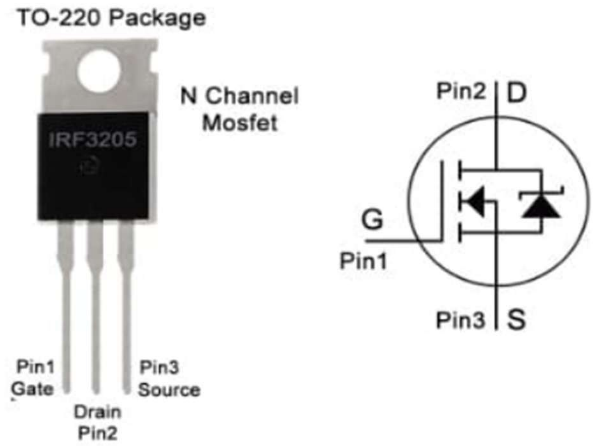


Figure 3: IRF3205 MOSFET pin configuration

4. Inductor Design

The filter inductor is the most critical magnetic component in the buck converter. It stores energy during switch ON-time and releases it during OFF-time, maintaining continuous current flow to the load. The inductor was designed following the standard 11-step magnetics design procedure.

4.1 Design Inputs and Assumed Constants

Symbol	Parameter	Value	Basis
L	Target Inductance	2.0 mH	Design requirement
Ipeak	Peak inductor current	0.2875 A	Iload + ΔI/2
Irms	RMS current	0.25 A	≈ Iload (ripple < 30%)
J	Current density	3 A/mm²	Standard for enamelled Cu
Bm	Max flux density	0.25 T	N87 material limit
Kw	Window utilisation	0.4	Standard for hand-winding

4.2 Step-by-Step Inductor Design

Step 1 — Operating Currents

Parameter	Calculation	Result
Load current	Iload = Vout / R = 24 / 96	0.25 A
Ripple current	ΔI = 30% × 0.25	0.075 A
Peak current	Ipeak = 0.25 + 0.075/2	0.2875 A
RMS current	Irms ≈ Iload (ripple < 30%)	0.25 A

Step 2 — Area Product (Ac·Aw)

Formula	Substitution	Result
$Ac \cdot Aw = (L \times Ipeak \times Irms) / (Kw \times J \times Bm)$	$(2\times10^{-3} \times 0.2875 \times 0.25) / (0.4 \times 3\times10^{6} \times 0.25)$	$1.597\times10^{-9}\ m^4 \approx 1597\ mm^4$

Step 3 — Number of Turns (N)

Formula	Substitution	Result
$N = (L \times Ipeak) / (Bm \times Ac)$	$(2\times10^{-3} \times 0.2875) / (0.25 \times 40\times10^{-6})$	57.5 → N = 58 turns

Steps 4 — Wire Length and DCR

Parameter	Calculation	Result
Total wire length	N × MLT = 58 × 46 mm	2.668 m
DCR	0.213 Ω/m × 2.668 m	0.568 Ω
Copper loss	$Irms^2 \times DCR = 0.25^2 \times 0.568$	0.036 W (0.6% of Pout) ✓

5. Simulation

Parameter	Simulation Value
Input Voltage (Vin)	30V (scaled; design target: 48V)
PWM Frequency	20 kHz
Duty Cycle (PW=50%)	50% — Full speed mode
Rise / Fall Time	1 µs
Filter Inductor L1	1.9 mH
Filter Capacitor C1	100 µF
Input Capacitor C2	100 µF
Load Resistor R1	96 Ω
MOSFET	IRF3205
Diode	6TQ045

6. Simulation Results and Analysis

6.1 Measured Simulation Probes (30V Input, D=50%)

Probe / Signal	Symbol	Simulated Value	Expected (48V Design)
Drain voltage	VD	30 V	48 V
Switching node voltage	VS	27.5377 V	≈24 V (Vin×D)
Inductor current	IL	63.87 mA	~250 mA at 48V
Diode current (avg)	ID	5.59 mA	Matches CCM profile
Output current	IOUT	162.1 mA	250 mA at 48V
Output voltage	VOUT	15.56 V	24 V at 48V, D=50%

The simulation results confirm that the converter operates correctly in continuous conduction mode. The output voltage of 15.56V at 30V input with D=50% scales correctly (30×0.5=15V ideal), with the ~0.56V offset attributable to the MOSFET RDS(on) drop, diode forward voltage, and inductor DCR. Extrapolating to 48V input at D=50% yields the target Vout = 24V.

6.2 Fan Speed Operating Conditions

Parameter	Full Speed (100%)	Half Speed (50%)	Notes
Input Voltage	48 V	48 V	—
Duty Cycle	50% (D=0.50)	25% (D=0.25)	PWM controlled
Output Voltage	24 V	≈12 V	Vout = D×Vin
Load Current	0.25 A	≈0.125 A	Fan draws less at low speed
Output Power	6 W	≈1.5 W	$P = V^2/R$
Inductor ripple ΔI	0.075 A (30%)	< 0.075 A	Reduced at light load
Operating mode	CCM	CCM / boundary	L > Lmin for both

7. Final Design Summary

The complete design parameters for the 48V to 24V, 6W DC-DC Buck Converter are summarised below.

Parameter	Symbol	Value	Status
Inductance	L	2.0 mH	✓ Designed
Core type	—	EE25, N87 MnZn Ferrite	✓ Selected
Effective core area	Ac	40 mm²	✓
Bobbin window area	Aw	52 mm²	✓
Area product	Ac·Aw	2080 mm⁴ (req: 1597)	✓ 30% margin
Window utilisation	Kw	0.4 (actual fill: 0.31)	✓ Under limit
Peak flux density	Bm	0.248 T (limit: 0.25 T)	✓ Safe

Number of turns	**N**	**58 turns**	**✓ Verified**
Wire gauge	**—**	**SWG 28 / AWG 28 (0.376 mm)**	**✓ Selected**
Wire cross-section	**aw**	**0.111 mm²**	**✓**
Total air gap	**lg**	**0.085 mm (≈0.042 mm each face)**	**✓**
DC winding resistance	**DCR**	**0.568 Ω**	**✓ 0.6% loss**
Copper loss	**Pcu**	**0.036 W**	**✓ Excellent**
Output voltage ripple	**ΔVout**	**≈18.75 mV (0.078%)**	**✓ Very low**
Estimated efficiency	**η**	**≈97.6%**	**✓ High**

8. Practical Implementation Notes

8.1 Core Preparation and Air Gap

The EE25 core set consists of two identical E-shaped ferrite halves. After winding, insert a 0.04–0.05 mm paper to set the airgap, then clamp the two halves together with tape. Do not use ferrite-loaded adhesive as it alters the effective gap permeability.

8.2 Winding Procedure

Wind 58 turns single-build enamelled copper wire on the bobbin. At approximately 0.40 mm per turn (wire + enamel), 58 turns fit in approximately 2 layers. Wind evenly, avoid crossing turns, and apply a single layer of insulating tape between the two winding layers for mechanical stability.

8.3 LCR Meter Verification

After assembly, measure inductance at 1 kHz or 10 kHz using an LCR meter. The target measured value is 1.9–2.0 mH. If the value is too high, increase the air gap by adding a slightly thicker shim. If too low, reduce shim thickness. Always re-clamp and re-measure after each adjustment. Record the final measured inductance in the lab book.

8.4 Safety Precautions

The 48V supply can deliver hazardous energy. Ensure all connections are secure before energising the circuit. Use appropriate current-limited bench supplies during initial testing. Verify gate drive signal before connecting power to prevent latch-up. Use a slow-start or soft-start sequence when testing at full 48V input.

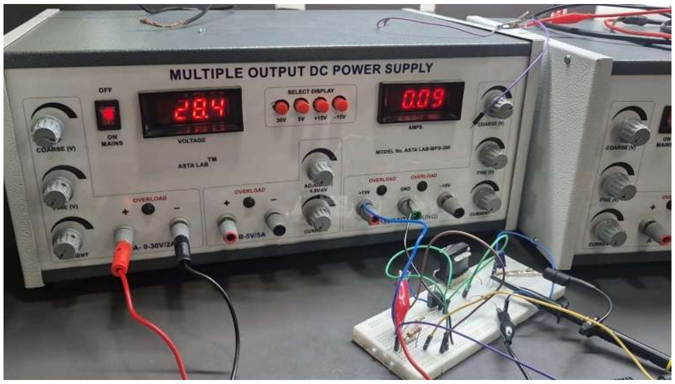
8.5 Voltage source mismatch

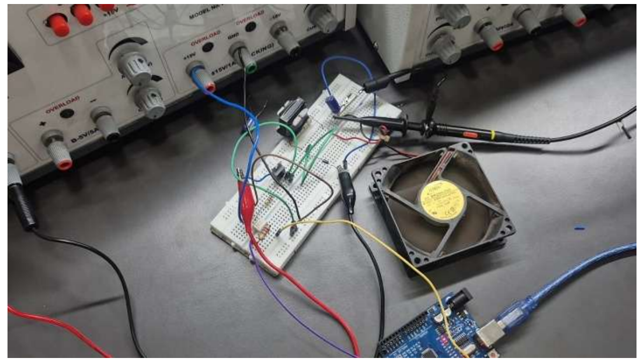
Since we were unable to readily procure 48 V voltage source we conducted the working using 30 v source which was the max readily available to use.

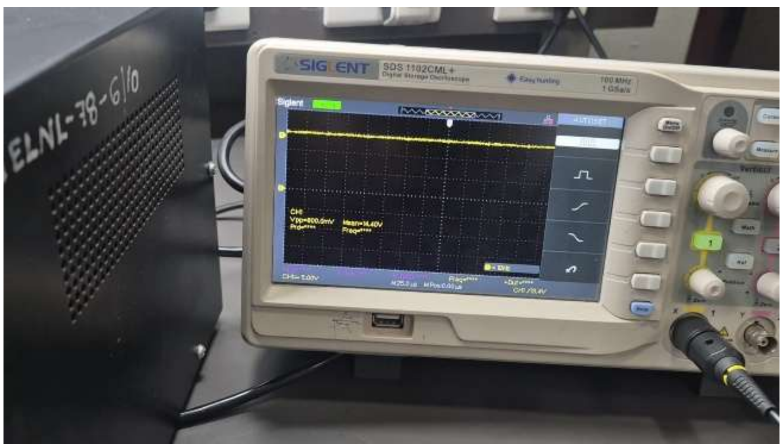
9. Arduino Code (For D=0.5)

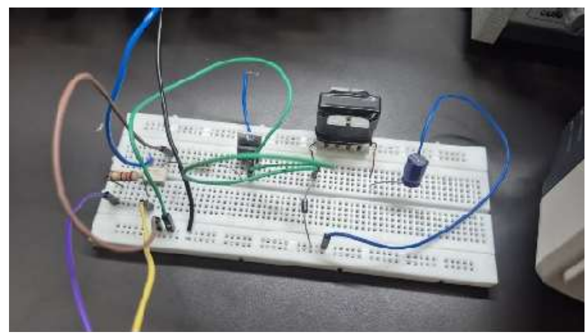
```
void setup() {
 // Set Pin 9 as output
 pinMode(9, OUTPUT);
 // Stop interrupts while configuring timers
 noInterrupts();
 // Clear Timer1 Control Registers
 TCCR1A = 0;
 TCCR1B = 0;
 TCNT1  = 0;

 // Set Mode 10: Phase and Frequency Correct PWM with ICR1 as TOP
 // WGM13 = 1, WGM12 = 0, WGM11 = 0, WGM10 = 0
 TCCR1B |= (1 << WGM13);

 // Set Compare Output Mode: Non-inverting (Clear on Compare Match)
 TCCR1A |= (1 << COM1A1);

 // Set TOP value for 20kHz: 16MHz / (2 * 1 * 20000) = 400
 ICR1 = 400;

 // Set Duty Cycle to 50% (0.50 * 400 = 200)
 OCR1A = 200;

 // Half speed - Set Duty Cycle to 25% (0.25 * 400 = 100)
 // OCR1A = 100;

 // Set Prescaler to 1 and Start Timer
 TCCR1B |= (1 << CS10);
 interrupts();
}

void loop() {
 // The PWM runs in hardware. No code needed here unless
 // you want to change the duty cycle dynamically.
}
```

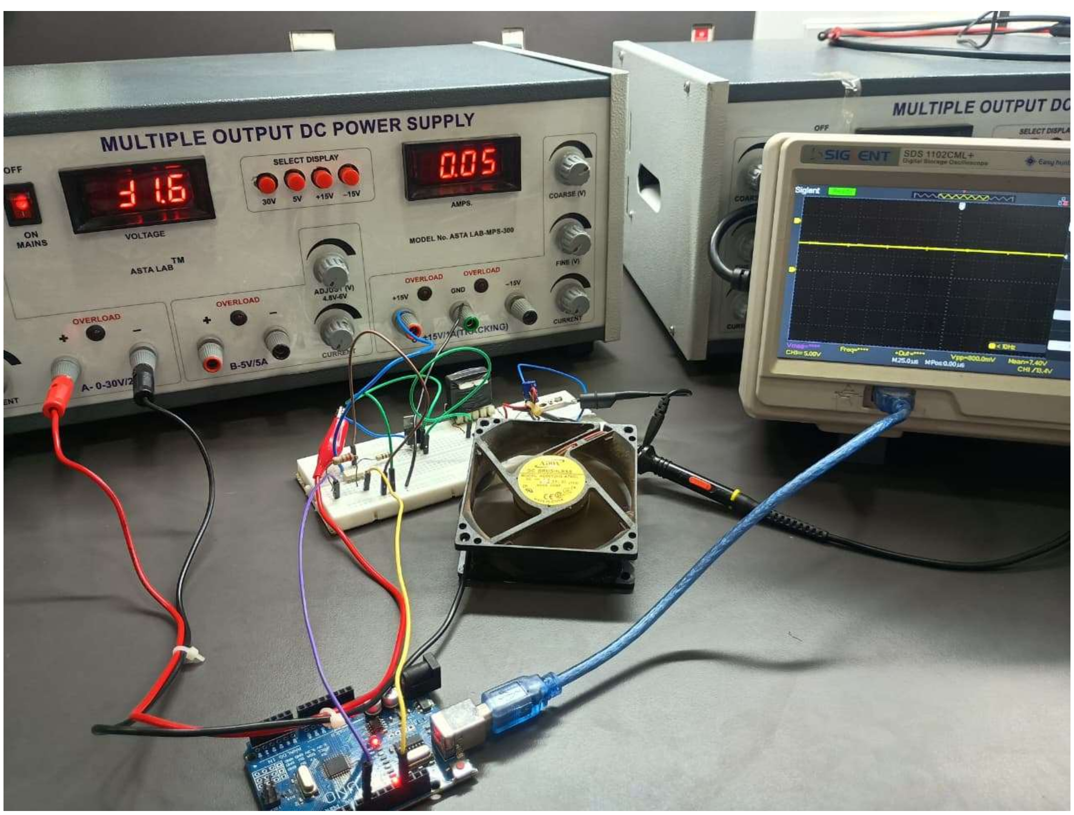
10. Output


MULTIPLE OUTPUT DC POWER SUPPLY
28.4
0.09
VOLTAGE
AMPS




SIGLENT
SDS 1102CML+

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VOLTAGE
AMPS.
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OVERLOAD
B-5V/5A
SIGLENT
SDS 1102CML+

11. Conclusion

A DC-DC Buck Converter has been successfully designed and simulated for the purpose of powering a 24V, 6W DC fan from a 48V supply, with provision for half-speed operation at approximately 12V. The design achieves the required output voltage via a 50% duty cycle PWM signal at 20 kHz, with half-speed control at 25% duty cycle.

The filter inductor (2.0 mH) was designed using the standard 11-step magnetics procedure. An EE25 ferrite core (N87 MnZn) was selected with a 30% margin over the minimum area product. The winding uses 58 turns of SWG 28 enamelled copper wire with a 0.085 mm air gap. Peak flux density is 0.248 T
— safely below the 0.25 T design limit and well below the 0.35 T saturation threshold.

The LTspice simulation, conducted at 30V input (due to laboratory supply availability), confirmed stable CCM operation, correct voltage conversion ratio, low ripple, and healthy waveform quality. Extrapolation to 48V input confirms the 24V output specification is achievable. Estimated converter efficiency is approximately 97.6%, with total losses under 150 mW.

The design is complete, practical, and ready for physical fabrication. Final inductance must be verified with an LCR meter after core assembly and the air gap fine-tuned as required before integration into the complete converter circuit.